Solution Requirements

Date	01 July 2026
Project Name	**Credit Card Approval Prediction**
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Users (Applicants/Admin) can securely log in using a username and password before accessing the system.*	High
2	Authorization levels	*Applicants can submit and view predictions. Admin can manage the dataset, trained model, and prediction records.*	High
3	External interfaces	*Integrates with Flask Web Application and IBM Watson Machine Learning for cloud deployment.*	High
4	Transactions processing	*The system accepts applicant details, preprocesses the data, performs prediction using the ML model, and displays the approval/rejection result instantly.*	High
5	Reporting	*Generate prediction history, model performance metrics (Accuracy, Precision, Recall, F1-Score), and downloadable reports*	Medium

6	Business rules, etc.	*Approval prediction is based on applicant income, employment status, credit history, loan status, and other financial attributes using trained ML models.*	High
7	Compliance to laws or regulations	*Protect user information according to data privacy principles and ensure confidential handling of applicant data.*	Medium
8	*Other*	*Provide an intuitive web interfac visualization.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	*The system should process and return prediction results quickly.*	*Prediction generated in **< 3 seconds***
2	Scalability	*The application should support increasing users and future model enhancements.*	*Supports **1,000+ prediction requests/day***
3	Security & Data Privacy	*Protect applicant information using secure communication and encrypted storage where applicable.*	*HTTPS enabled, encrypted data storage, secure login)*

4	Reliability & Availability	*The application should remain available with minimal downtime.*	*99% system availability*
5	Usability & Accessibility	*The web interface should be easy to use, responsive, and accessible on desktop and mobile devices.*	*User-friendly interface with responsive design*
6	*Other*	*The system should be modular, easy to maintain, and deployable on different operating systems or cloud platforms.*	Supports Windows, Linux, macOS, and IBM Cloud deployment